

Qualified Aerospace Instructors Course

From Surveillance to Strike: The Involvement of Drones in the Defence of Ukraine

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Contents:

Section	Page
Introduction	3
Historical Context	3
Drones Used by Ukraine	4
Primary Benefits	5
Notable Challenges	6
Conclusion	6
Bibliography	7

Introduction

Drones are at centre stage of the war in Ukraine. From targeting exposed mortar positions, to collecting invaluable intelligence about the battlefield environment, to even assisting with the sinking of the *Moskva*, they have allowed Ukraine to offer stiff resistance to its invaders. The sheer numbers of drones that Ukraine possess cannot be overstated; Ukraine deployed an estimated 200,000 drones in 2023,^[1] varying from the largest Unmanned Aerial Vehicles (UAVs) exceeding 450kg, to ones small enough to fit in the palm of one's hand, and will produce one million drones over the course of this year, according to Ukrainian President Volodymyr Zelenskyy.^[2]

However, drones no longer hold the same 'game changer' status they held at the outbreak of the conflict, with Ukrainian UAV losses standing at 10,000 *per month*, according to the Royal United Services Institute.^[3] Additionally, Russian forces are now deploying drones of their own, as well as finding new ways to counter Ukraine's drone fleet, forcing a shift in tactics from the defenders.

This essay is intended to analyse how the Ukrainian Armed Forces has changed its strategy to adapt to these developments, and the benefits to other countries' armed forces of an increased use of drones.

Historical Context

After the 1991 dissolution of the Soviet Union, which both Russia and Ukraine were a part of, the two newly formed countries maintained close ties into the 21st century. However, beginning in 2004 with the Orange Revolution, and continuing in 2014 with the Euromaidan movement, Ukraine began to politically distance itself from Russia, in favour of building ties with the EU, NATO, and the West.^[4]

After the Euromaidan movement forced pro-Russian President Viktor Yanukovych out of office, the resulting power vacuum and civil unrest in the Russian-speaking Eastern regions encouraged the Kremlin to invade and annex the Crimean Peninsula, leading to general unrest in the Eastern Donbas region from 2014. This led to a regional conflict between Ukraine and Russian-backed separatist movements in the East, known as the War in Donbas, which is considered to be the start of the wider Russo-Ukrainian War.

It was during this conflict that drone technology became prevalent, mostly by the Russians using UAVs to hunt for Ukrainian forces, which encouraged Kyiv to develop its own drone technology.^[5] As a result, the 2022 Russian Invasion of Ukraine had an initial drone capability gap in Ukraine's favour, ultimately contributing to a near-catastrophic initial invasion attempt by the Kremlin.^[6]

Drones Used by Ukraine

Bayraktar TB2

During the initial stages of the war, the most prominent drone in Ukraine's arsenal was the Turkish Bayraktar TB2, at a cost of approximately 5 million USD per unit. Capable of fully autonomous flight, it flourished in the early battlefield environment where Russian forces were continually in exposed positions, and Russian air defence and electronic warfare capabilities were severely lacking. Being able to loiter for up to 25 hours, with a range of up to 300km, and the ability to carry four laser-guided munitions,^[7] the TB2 is a formidable weapon against an enemy without the means to defend itself against it.

Due to this favourable battlefield environment, the TB2's primary role revolved around ground attack, particularly for the purpose of Suppression of Enemy Air Defences (SEAD), Air Interdiction, and deep attack. Initially supplied with 20 TB2s, Ukrainian forces managed to destroy an impressive number of strategic assets, including Buk and Tor anti-air missile systems,^[8] several T-72 tanks, self-propelled guns, infantry fighting vehicles,^[9] and an Mi-8 helicopter.^[10] Perhaps most notably, it was drones, likely TB2s, that distracted the Russian Cruiser *Moskva's* defences away from the Neptune missiles that sunk it, resulting in Russian naval forces being forced to displace from the Black Sea, withdrawing their air defence networks with them, as well as eliminating the possibility of a Russian amphibious assault in Odesa.^[11]

However, with Russian air defence systems now becoming far more robust and dependable, the size of the TB2 means it is far less capable of sneaking into Russian-held territory and inflicting considerable damage without being detected and intercepted, particularly as Russia gains a favourable air situation. Furthermore, their large cost means that they have become far less cost-effective compared to other drone systems. As a result, their primary role has shifted, making use of their advanced sensor suite for the purposes of ISR, allowing smaller, more tactical drones to strike at valuable targets instead.^[12]

AeroVironment Switchblade

Developed by the US armed forces to provide rapid air support in the rugged terrain of Afghanistan, the AeroVironment Switchblade has also found moderate success in Ukraine. Coming in two variants (the smaller, anti-personnel Switchblade 300, and the heavier, anti-armour Switchblade 600), the Switchblade is effectively a mix between a flying scout and a guided weapon, formally known as a '*loitering munition*.'^[13] Effectively, it flies around until it locates a target, either via its own sensors or from targeting information from other drones, at which point it will fly down to the target and detonate next to it, inflicting considerable damage.

From Surveillance to Strike: The Involvement of Drones in the Defence of Ukraine

Although Switchblades had initial success, as Russian air defence systems, especially radio jammers, became more robust, its usage gradually declined. In July 2023, the US Army stated it would not purchase any more Switchblade 300s to supply to Ukraine, due to their prohibitive cost (around 6,00 USD/drone, not including their control systems) and inferior performance against armoured vehicles.^[14]

Off-the-Shelf Drones & Donated Drones (*'Dronations'*)

Due to the limitations of the aforementioned high-tech yet extremely costly drones, Ukraine has transitioned to relying on smaller, often modified commercial drones to shape the battlespace. Many first-person view (FPV) drones are equipped with explosive devices, such as grenades, and flown towards targets, similarly to the Switchblade. However, these smaller, cheaper, and more replaceable drones are far easier to use, and far less detectable by Russian radar systems.

To support this effort, the Ukrainian domestic drone market has boomed over the past year, rising from just seven companies producing drones to at least eighty.^[15] Additionally, many hobbyists and workshops are dedicated to the task of transforming off-the-shelf drones into instruments of war. Some drones made by these hobbyists cost as little as \$10 per unit. Meanwhile, the so-called *'Eyes of Army'* in Kyiv is specialised in converting eight-rotor drones into "heavy" bombers, carrying multiple 3kg mortar rounds, which, according to the Economist, is enough to destroy a tank.^[16]

Although these drone systems are more able to overwhelm some types of air defences, like SAM sites, they are more susceptible to Russian jamming systems due to a lack of advanced communication systems and increased vulnerability to jamming efforts, rendering them completely unusable around areas with robust Russian radio jamming systems.

Primary Benefits

The use of drones by Ukraine is responsible for benefits to the Ukrainian armed forces that are near impossible to overstate. The real-time intelligence provided by drones allows for enhanced decision-making and operational planning, based off real-time troop movements, battlefield conditions, and the concentration of enemy forces.

Additionally, the tactical intelligence that can be gained from drones with long loiter times allow for cheap, replaceable swarms of drones to attack an area by overwhelming air defences and dropping explosives at the target, guided by the loitering drone (commonly the TB2). This will reduce collateral damage, since there will be a smaller blast radius from a drone-mounted explosive as opposed to a

missile, and allows for better conservation of resources, since the observer drone can be re-used on another mission.

Drones also allow for a shortened 'kill-chain,' shortening the time between target acquisition and target destruction,^[15] whilst also allowing for deep strikes to be conducted well into Russian territory, such as the Engels air base strikes, and the more recent drone strikes around St Petersburg.^[17] Even when shot down they are extremely cost-effective, since a \$200,000 Buk missile is a lot more difficult to replace than a \$200 modified drone from a shop shelf.

Finally, they provide a huge adverse psychological impact on Russian forces. The sinking of the *Moskva* would have undoubtedly made numerous Russian soldiers uneasy enough; to know that a drone was partially responsible for its loss would have no doubt caused widespread anxiety within the ranks.

Notable Challenges

Although drones have had a significant impact on the battlespace in Ukraine's favour, there are disadvantages to this increased reliance.

Due to Russia's air defence network becoming more capable, it is now more adept at detecting larger drones, meaning that drones such as the TB2 cannot be fully utilised. Additionally, an increased prevalence of radio jammers, spoofers, and lasers has resulted in Ukrainian drones becoming more vulnerable, with losses standing around 100,000 per month.

Most importantly, the conflict currently stands at a stalemate, with Russian forces digging in to fortified positions. Ultimately, this has reduced the effectiveness of the drone fleet since there is less opportunity to target exposed forces from above.

Conclusion

In conclusion, it is clear that Ukraine has greatly benefitted from its drone fleet. The ability to provide ISTAR capabilities, conduct airstrikes, and demoralise the enemy, all without having to risk high-value aircraft is irreplaceable. These drones represent strategic foresight, adaptability, innovation, and resilience in the face of adversity.

Ukraine is using its drones not only as flying machines, but as force multipliers. The rest of the world should take notice and follow suit.

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From Surveillance to Strike: The Involvement of Drones in the Defence of Ukraine

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